Write a Python program to implement a Convolutional Autoencoder using the CIFAR-10 and Fashion-MNIST image datasets. Load both datasets and perform exploratory analysis to understand their image dimensions, number of classes, number of samples, and pixel-value distributions. Preprocess the images by appropriately resizing them, normalizing pixel values, and converting them into a suitable format for input to a convolutional neural network. Design an Encoder–Decoder architecture in which the encoder uses convolutional and pooling layers to transform the input image into a lower-dimensional latent-space representation, while the decoder uses upsampling/transposed convolution layers to reconstruct the original image. Train the Autoencoder separately on the CIFAR-10 and Fashion-MNIST datasets, using the original images as both the input and reconstruction target. Experiment with different latent-space dimensions, such as 16, 32, 64, and 128, and compare their reconstruction performance. Also calculate the error rate for the reconstructed image.

CIFAR-10: https://www.kaggle.com/code/farzadnekouei/cifar-10-image-classification-with-cnn

Fashion-MNIST: https://www.kaggle.com/code/hossamahmedsalah/fashion-mnist